Vadthyavath Venkatesh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.47
Narayana JR college	Class XII (Telangana board of intermediate education)	2024	97.60%
TSWREIS Boys High School	Class X (board of secondary education)	2022	9.80%

SKILLS

Data & Analytics: Advanced Excel, Data Analysis, Statistical Analysis, Data Visualization, Demand Estimation, Basic SQL

Programming: Basic Python, Pandas, NumPy

Project Management: Project Planning, Project Coordination, Requirement Analysis, Stakeholder Management, Resource Planning, Risk Management, Process Improvement

Tools: Microsoft Excel, PowerPoint, Jira, MS Project, Trello

POSITIONS OF RESPONSIBILITY

Hostel Representative (HR), IIT Hyderabad

- Represented a 300+ resident community, coordinating with students and the administration to resolve concerns, improve communication, organize hostel-wide events, and support student welfare initiatives.

Head of Ideation & Operations, Alumni Cell - IIT Hyderabad

- Led ideation-to-execution for alumni engagement programs, coordinating cross-functional teams and optimizing workflows to ensure successful delivery of alumni networking events.

EBSB Events Coordinator, Ek Bharat Shreshtha Bharat (EBSB)

- Coordinated institute-wide events with multidisciplinary teams, managing stakeholder engagement and end-to-end execution to drive seamless cultural exchange programs.

Sunshine Mentor, Sunshine - Counselling Cell

- Mentored incoming undergraduate students, facilitating their academic, personal, and campus transition through structured peer support.
- Fostered student well-being and engagement by providing guidance and promoting effective integration into the institute community.

PROJECTS

Ride-Hailing Marketplace Optimization | Product Case Study

- Conducted **Problem Discovery & User Research** to analyze post-acceptance ride cancellations and identify rider, driver, and marketplace pain points.
- Performed **Root Cause Analysis (RCA)** to identify key cancellation drivers across **rider, driver, and marketplace** factors.
- Developed **Guesstimates** to estimate cancellation scale and its impact on **rider wait time, driver utilization, and marketplace efficiency**.
- Evaluated **Product Improvements** to reduce cancellations while maintaining **driver economics and rider**

availability.
- Defined **Product Metrics & Guardrails** including completion rate, cancellation rate, wait time, driver acceptance, earnings/hour, utilization, and churn, with an **A/B Testing** approach.

ChemFlow AI - Chemical Engineering Process Simulation Platform | Live Demo

- Developed a **Material Balance Calculator** to perform steady-state mass balance calculations for multi-stream chemical processes.
- Built an **Energy Balance Calculator** to analyze heat transfer, energy conservation, and process energy requirements.
- Implemented a **Reactor Simulator** supporting batch, CSTR, and PFR models for reaction engineering analysis.
- Designed a **Heat Exchanger Calculator** for evaluating heat transfer performance and thermal design parameters.
- Created a **Distillation Column Calculator** to assist in separation process analysis and column design calculations.
- Developed a **Process Flow Designer** for creating and visualizing chemical process flow diagrams through an interactive interface.
- Integrated a **Chemical Database**, engineering graphs, and an **AI Process Advisor** to provide process insights and improve engineering decision-making.

RELEVANT COURSES

Relevant Coursework: Design Thinking for Innovation, Marketing Research, Introduction to Entrepreneurship, Communication Skills

EXPERIENCE

1. CSIR-IICT | Research Intern -- Worked on **two government projects** under a scientist, gaining **hands-on experience** in project execution, data analysis, and problem solving.

2. IIT Hyderabad | Research Project -- Worked on **building a spectroscopy system**, involving requirement analysis, testing, troubleshooting, and optimization.

EXTRACURRICULAR

State-Level Finalist, Super Student Competition, Telangana Social Welfare Residential Educational Institutions Society (TSWREIS)

- Advanced to the state level in a prestigious teaching competition through outstanding performance in subject expertise, teaching methodology, communication, and classroom presentation skills.